

Parth Mital

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Education

Vellore Institute of Technology (VIT)
B.Tech in Computer Science & Engineering
CGPA: 8.70 / 10

Vellore, India
2023 – 2027

Projects

NeuroScan AI - Medical Imaging Research & Full-Stack Analysis Platform

🐙 github.com/parthmital/NeuroScan-AI

- Benchmarked **21 deep learning architectures** on **9,000+ MRI scans** for automated brain tumour analysis.
- **Detection: 99.59% accuracy** using CoaTNet-0, outperforming ResNet/Inception baselines.
- **Classification: 98.58% accuracy**; reduced dataset noise by **19%** via perceptual hashing **deduplication**.
- **3D Segmentation:** 3D U-Net achieved Dice scores of **0.847 (WT)**, **0.753 (TC)**, **0.761 (ET)**.
- Built **clinical SaaS platform** with **JWT authentication**, **NIfTI pipelines**, & **vtk.js 3D visualization**.
- **Tech:** Python, PyTorch, TensorFlow, FastAPI, React 18, TypeScript, SQLModel, Tailwind, vtk.js, OpenCV.

VIT Carpool - Full-Stack Campus Ride-Sharing Platform

🔗 vitcarpool.vercel.app 🐙 github.com/parthmital/VIT-Carpool

- Built **centralized ride-sharing platform** replacing fragmented WhatsApp-based coordination.
- Implemented **structured filtering system** (source, destination, time) for **real-time** efficient ride discovery.
- Achieved **2,600+ views/impressions** within the VIT student community, indicating strong engagement.
- Secured access via domain-restricted **Google OAuth** & enforced data isolation using **Supabase RLS**.
- Developed **responsive UI** using React 18, Tailwind CSS, Radix UI, & Framer Motion.
- **Tech:** React 18, TypeScript, Vite, Tailwind CSS, Supabase, PostgreSQL, REST APIs, Google OAuth.

VIT Resource Automation Pipeline - Academic Resource Lifecycle Toolkit

🐙 github.com/parthmital/VIT-Resource-Automation-Pipeline

- Built **end-to-end automation pipeline** processing **6,000+ academic documents** per semester, reducing collection time from **weeks** to under **1 hour**.
- Developed asynchronous Selenium/Playwright scrapers for **PYQs**, reducing per-subject effort from **minutes** to **seconds**.
- Automated **bulk data extraction** from VTOP/UniBud across **150+ faculty** & **6 concurrent subjects**.
- Implemented **analysis engine** (pypdf + COM) to rank faculty via slide count & content depth.
- **Standardized** legacy files (.doc, .ppt) into 100% PDF format for consistent access.
- **Tech:** Python, Selenium, Playwright, pypdf, pywin32, comtypes, Chrome DevTools Protocol, CSV.

Technical Skills

Languages: Python, C++, Java, TypeScript, JavaScript

Core Computer Science: Data Structures & Algorithms, Operating Systems, Computer Networks

Machine Learning & Computer Vision: PyTorch, TensorFlow, CNNs, Transformers, OpenCV

Backend & Databases: FastAPI, REST APIs, PostgreSQL, Supabase, MongoDB, JWT Authentication

Frontend: React, Next.js, Tailwind CSS

Cloud & DevOps: AWS, Cloud Architecture, Docker

Automation & Systems: Selenium, Playwright, Web Scraping, Data Pipelines, ETL, Automation

Systems & Hardware: Microprocessors & Microcontrollers, Digital Systems

Tools: Git, Vite, vtk.js, NIfTI Processing, pypdf, pywin32, comtypes

Experience

Web Development Intern - Hiraeth Tech

Jun 2025 – Jul 2025

Vadodara, India

- Built **5+ responsive WordPress websites** using Elementor and custom CSS.
- Optimized images and assets to improve page load performance.
- Ensured cross-device responsiveness and consistent UI/UX.